

Full Length Article

Fast breathing facilitates reaction time and movement time of a memory-guided force pulse

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ABSTRACT

Slow controlled breathing can be beneficial for performance of continuous and serial motor tasks. However, how controlled breathing influences discrete motor task performance remains unclear. We sought to determine the impact of paced breathing frequency on measures of movement initiation (reaction time: RT), accuracy (absolute endpoint error: AE; constant error: CE), and variability (trial-to-trial variability: V), in a goal-directed discrete motor task. We hypothesized slow breathing would be accompanied by faster RT, reduced AE and CE, and less V compared to faster breathing rates. Participants ($N = 47$) performed a memory-guided force pulse pinch task targeted at 10% of their maximum voluntary contraction while breathing at metronome-paced slow, normal, and fast frequencies. During each breathing condition, heart rate variability (HRV) as indexed by the standard deviation of 'NN' intervals (SDNN) was measured to ensure objective manipulation check of participants breathing at their set pace. Following each breathing condition, participants provided subjective ratings using the Affect Grid and Visual Analog Scales for arousal, hindrance, and dyspnea. Manipulation check results indicated participants correctly breathed at metronome paces, as indexed by increased HRV for slow breathing and decreased HRV for fast breathing. Results indicated that fast breathing reduced reaction time and movement time, and increased ratings of arousal, hindrance, and dyspnea. In contrast, slow breathing increased reaction time, and levels of hindrance and dyspnea were similar to normal breathing. Breathing frequency did not differentially impact accuracy or variability across conditions. Findings provide evidence that breathing frequency affects fundamental movement parameters, potentially mediated by factors other than arousal.

1. Introduction

Successful motor performance is dependent upon fulfilling context specific demands for accuracy, consistency, and speed (Schmidt, Zelaznik, Hawkins, Frank, & Quinn Jr., 1979). All three components are essential to effectively executing fine motor tasks in low and high-risk contexts (Ritter & Scott, 2007; van Riet-Nales et al., 2014). Failure to perform effectively within low and high-risk contexts compromises performance outcomes, as well as injury or mortality due to deviations within and amongst movement parameters that underlie overt motor actions (Hussain, Latif, & Choudhary, 1988). A need exists to identify low cost strategies to maximize movement effectiveness and efficiency. Breathing interventions hold great promise for fulfillment of this need (Rivest-Gadbois & Boudrias, 2019).

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Accordingly, evidence exists to support the notion that controlled breathing frequency can influence motor performance, but the mechanisms underlying these effects remain unspecified (Singh & Mutreja, 2020; Zaccaro et al., 2018).

Breathing interventions manipulate respiratory parameters to alter physiological mechanisms that contribute to performance (Song & Lehrer, 2003). Specifically, reducing breathing frequency increases heart rate variability (HRV), which improves broad performance outcomes in athlete populations (Morgan & Mora, 2017). For example, reducing breathing frequency has been shown to increase HRV and improve technique in professional dancers (Raymond, Sajid, Parkinson, & Gruzelier, 2005), as well as improve dribbling, shooting, and passing skills in basketball players (Paul & Garg, 2012). Although these studies provide affirmation that breathing interventions facilitate execution of continuous and serial movements, they do not address *how* breathing frequency influences core components of motor planning and execution.

The corpus of literature addressing the impact of breathing interventions on performance is replete with limitations and unfounded assumptions. Specifically, questions remain unanswered regarding which mechanical components of breathing are most effective. We sought to alleviate some of the inferential gaps in the current literature base by focusing on fundamental questions concerning the influence of breathing frequency on a motor planning dominant task. Our primary aim was to determine whether and how breathing frequency affects various indices of motor planning. More specifically, the purpose of the current study was to determine the impact of breathing frequency on a goal-directed discrete motor task emphasizing accuracy.

The memory-guided force pulse (a discrete pinch) task featured in the current investigation is a simple yet elegant task which manifests overt representations of the cognitive and motor processes involved with movement planning (Coombes, Higgins, Gamble, Cauraugh, & Janelle, 2009). Planning-dependent movements involve selecting an appropriate motor program, while control-dependent tasks allow time to monitor and adjust the movement during execution. The pinch task used herein required fast initiation. A speed-oriented goal renders the movement heavily dependent on the motor planning system. Performing the task as instructed (i.e., quickly) prevents control-based adjustments to be made as the response unfolded, thereby allowing inferences to be made concerning the impact of breathing frequency on motor planning (Glover, 2004). Furthermore, previous evidence using TMS stimulation on motor planning areas of the brain demonstrates that force outcomes of planning-dependent movements (such as reaction time) are facilitated (Coombes, Tandonnet, et al., 2009).

Previous studies on breathing and motor performance include slow breathing frequency manipulations, while excluding fast breathing frequencies (Kim & Tennant, 1993). Whether the espoused benefits of breathing on motor tasks are due to breathing frequency or some other attribute of breathing remains unclear. The effect of slow, fast, and resting paced breathing frequencies on various components of performance, including motor *accuracy*, *consistency*, and *speed* remains largely unknown. Herein, motor accuracy was assessed by absolute endpoint error (AE) and constant endpoint error (CE), motor consistency was assessed by the trial-to-trial variability of force (V), and movement imitation was assessed by reaction time (RT), peak rate of force (PRF), and normalized movement time (NMT). The rationale for including these measures is summarized below, as are the expectations associated with them in the current protocol.

No studies to date have examined the influence of breathing frequency on the accuracy of motor performance, but the influence of breathing frequency on cognitive performance has been studied. Previous research reports that slow breathing frequencies increase accuracy (decrease error) of modified Stroop task compared to normal breathing frequencies (Prinsloo et al., 2011). Cognitive performance is essential for movement planning of novel skills prior to those movements becoming automatic. Therefore, we expected slow breathing to yield decreases in error during execution of novel planning-dependent motor actions. In addition to testing this prediction, we also sought to determine the impact of other breathing frequencies (i.e., fast-paced breathing) relative to resting state (normal) breathing. Based on the previously mentioned studies, we assumed that any faster frequencies that deviate from slow breathing frequencies might also increase error. Furthermore, we expected fast breathing to increase performance error, and slow breathing to decrease performance error.

Consistency is a measure of variability, referring to the standard deviation of the observed performance from the target goal (Schmidt et al., 1979). Albeit limited, empirical efforts have been made to investigate how breathing frequency impacts variability of a *sustained* force production task. For example, rapid inspirations and expirations increase variability during a sustained isometric task (Li & Yasuda, 2007). Greater attention allocation has also been shown to increase motor variability under dual task situations (Dubost et al., 2006). Although the impact of slow breathing on motor variability is unknown, indirect evidence described above from studies of cognitive performance allows strong inference that slow breathing is comparatively less attention demanding (Prinsloo et al., 2011). Accordingly, we expected fast breathing to increase motor variability and slow breathing to decrease motor variability relative to normal breathing.

Controlling breathing frequency is more cognitively demanding than passively breathing, which is more automatic (Doll et al., 2016). Specifically, attention to one's breath activates brain regions associated with attentional control (Dickenson, Berkman, Arch, & Lieberman, 2013). Taking into account attentional demand of controlled breathing, we were interested in the impact of breathing frequency alone. We therefore included a controlled normal breathing frequency condition. In this condition, participants breathed at the same frequency as they did when passively breathing, but they were required to breathe with external pace of a metronome.

Reaction time is a fundamental component of response speed, referring to the onset of movement in response to an external stimulus (Galaro, Celnik, & Chib, 2019). Reaction time is measured verbally, auditorily, or behaviorally to inform planning and decision-making processes (Jose & Gideon Praveen, 2010). While verbal and auditory reaction time index decision-making, behavioral measures of reaction time index motor planning processes (Wasson, Prodoehl, Coombes, Corcos, & Vaillancourt, 2010). Various factors alter motor planning, including, arousal. Individuals employ overt or covert regulation strategies on a daily basis to regulate arousal (Gross, 1998). Regulation strategies provide benefits and costs to motor planning depending on the strategy used. Slowed reaction times resulting from implementation of regulation strategies represent a disruption in the efficiency in motor planning (Beatty,

Fawver, Hancock, & Janelle, 2014). Alternatively, breathing interventions have the potential to facilitate motor planning. Long-term training of slow and fast breathing increases or sustains reaction speed (Madanmohan Udupa, Bhavanani, Vijayalakshmi, & Surendiran, 2005; Sharma, Subramanian, Bhavanani, Madanmohan, & D., 2014). Single session studies found decreases in reaction time during slow breathing conditions (Dhadse & Fadia, 2016; Prinsloo et al., 2011). Notably, while prior research assesses reaction time, the speed and accuracy of decision-making has been the primary focus in this prior work, with the ensuing movement being largely ignored. We sought to fill this gap in the literature by determining whether effects of breathing frequency extend beyond the initiation of an obligatory movement to affect other movement attributes. The protocol used in the current study required execution of a memory-guided force pulse in which accuracy was emphasized. Directing attention away from a task slows down reaction times (Jehu, Despons, Paquet, & Lajoie, 2015). Fast breathing may require comparatively greater attention to breathing itself, thereby diverting it from the task (Li & Yasuda, 2007). Consequently, we hypothesized fast breathing would increase reaction time while slow breathing was expected to decrease reaction time.

In addition to attention, arousal is an important potential contributor to alterations in performance measures. Slow breathing strategies decrease objective indices of arousal, whereas fast breathing increases objective indices of arousal (Martin et al., 2012), which can affect performance (Coombes, Higgins, et al., 2009). Accordingly, we also sought to determine whether breathing frequency influences subjective and objective indices of arousal. Arousal levels were assessed using self-report subjective questionnaires and standard deviation of NN intervals (SDNN) of HRV. We expected subjective ratings of arousal to mirror findings of objective arousal with participants rating slow breathing as less arousing compared to normal breathing conditions, and participants rating fast breathing as more arousing compared to normal breathing conditions. Subjective changes in arousal would provide confirmation and replication of previous findings that slow breathing decreases arousal and fast breathing increase arousal.

In sum, we predicted that slow breathing would decrease AE and CE (Paul & Garg, 2012) relative to fast and normal breathing. Increased error is associated with increases in variability as indexed by V (Enoka et al., 2003). Therefore, we also hypothesized that increased V would be evidenced in the fast breathing conditions relative to slow and normal breathing. As described, attention allocation is also a component of breathing. Implementation of strategies that impair cognitive processes such as attentional deployment would be expected to increase RT and PRF while decreasing NMT. Thus, we hypothesized fast breathing would be more cognitively demanding and increase RT and PRF, while decreasing NMT accordingly. Greater arousal is associated with faster response speed (Arent & Landers, 2003). We therefore explored the possibility that arousal increases would be greater during fast breathing, which might decrease RT in the fast breathing condition.

2. Methods

2.1. Participants

Fifty participants from a large university in the southeastern United States participated in this study and received extra course credit for their participation. We included participants who self-reported they were right-handed, as well as reported no movement, respiratory, emotional, or cardiovascular disorders. Each participant provided written informed consent prior to beginning the study. Exclusion of data from three participants occurred due to technical issues. Results are reported for the remaining forty-seven undergraduate students (30 women: mean [*M*] age = 20.5, standard deviation [*SD*] = 2.8; 17 men: *M* age = 22.1, *SD* = 2.8). Participants responded similarly to an assessment of dispositional characteristics, including the depression, anxiety, and stress scale (DASS), characterizing participants' level of related negative emotional states (Table 1). Scores were similar to normative values for the DASS, depression ≤ 9 , anxiety ≤ 7 , and stress ≤ 14 (Lovibond & Lovibond, 1995). In addition, males and females responded similarly across the self-report assessments. Table 1 also summarizes the demographic information for all participants included in the analyses.

2.2. Instrumentation and task

Participants sat with their head positioned one meter from a computer screen (23.6 in, 60 cm; 1920 × 1080 resolution) and performed isometric memory-guided force pulse contractions with their right hand by pinching a force transducer (MLP-75, Transducer Techniques, Temecula, CA, USA) with the thumb and index finger. The force transducer's vertical position was adjusted to allow participants' right arms to relax comfortably on the armrest at a right angle. Analog output from the force transducer was amplified through a 15LT Grass Technologies Physiodata Amplifier System (Astro-Med Inc. West Warwick, RI, USA) at an excitation voltage of

Table 1
Demographic data for the 47 participants (30 women) included in the statistical analyses.

Variable	All Participants (<i>N</i> = 47)	Male (<i>N</i> = 17)	Female (<i>N</i> = 30)
	<i>M</i> (<i>SD</i>)	<i>M</i> (<i>SD</i>)	<i>M</i> (<i>SD</i>)
Target force (N)	4.11 (1.43)	5.11 (1.)	3.56 (1.13)
DASS- Total	13.40 (15.83)	9.88 (11.03)	15.40 (17.85)
DASS-D	3.32 (6.83)	2.24 (2.99)	3.93 (8.25)
DASS-A	3.49 (4.44)	2.59 (3.06)	4 (5.04)
DASS-S	6.60 (6.90)	5.06 (6.05)	7.47 (7.30)
BMI	23.22 (3.26)	24.96 (3.50)	22.29 (2.58)

10 V. Custom LabVIEW software (14.0.1; National Instruments, Austin, TX) controlled trial onset, trial offset, visual stimulus presentation, and a 16-bit analog-to digital converter (A/D) (PCI-6220, National Instruments, Austin, TX) which sampled the force at 100 Hz. Force data were analyzed offline. Participants' breathing frequencies were recorded by a force transducer belt (Pneumotrace; Morro Bay, CA, USA) placed 2 cm above their umbilicus around their abdomen. We instructed participants to limit extraneous movement to reduce signal noise. Table 2 summarizes the average breathing frequencies during each breathing condition, as well as standard deviations. In addition to the force transducer belt, participants wore a Polar heart rate monitor belt H10 around their chest (in line with xiphoid process). Measures were recorded during each breathing condition.

2.3. Breathing manipulation

Participants performed slow-paced, normal-paced, and fast-paced breathing frequencies, each of which were paced by a smart-phone metronome app (Pro Metronome App., EUMLab, Hangzhou, China). Participants practiced with investigators to control accuracy of breathing pace with the metronome-pace (see Fig. 1). Normal-paced breathing involved breathing at the same breaths per minute that were established at baseline. A baseline condition was included to control for potential effects of attention to the metronome; subsequently, baseline performance outcomes were used in change scores. Slow-paced breathing involved altering breathing frequency to the pace exhibiting the largest increase in HRV, based on a previous protocol (Lehrer, Vaschillo, & Vaschillo, 2000). Specifically, participants breathed at their resonance frequency for slow-paced breathing, characterized as highest peak trough amplitude, highest LF, the maximum LF amplitude peak on the spectral graph, and cleanest and highest-amplitude LF peak as referred to by Lehrer and colleagues (Lehrer et al., 2013). Fast-paced breathing was set at an interval equivalent to the breaths per minute difference between slow- and normal-paced breathing. Prior to the current experiment, pilot testing of the fast breathing frequency ensured safely avoiding induction of hyperventilation. Randomized and counterbalanced breathing conditions comprised instructions informing participants to breathe in and out with a metronome. They had approximately two minutes to acclimate to the breathing frequency prior to beginning the experimental condition. Instructions focused on participants' ability to breathe in through their nose and out through their mouth to prevent dry mouth (Bianchi et al., 2004). Lastly, participants breathed abdominally to ensure the strain gauge belt recording breathing frequencies.

2.4. Experimental design and procedure

We employed a four level (condition: baseline, slow, normal, fast) repeated measures experimental design. To begin the experiment, individualized maximal voluntary contraction (MVC: in Newtons) values were calculated for each participant using a previously established protocol (Coombes, Higgins, et al., 2009; Vaillancourt & Newell, 2003). Participants then trained to achieve motor task competency during two automated pre-experimental training sessions, requiring participants to produce a pinch force equaling 10% of their individualized MVC. The training session concluded once participants were able to complete eight out of ten of the most recent consecutive trials within a range of $\pm 20\%$ of the individualized target force. Participants received feedback after each trial consisting of a force curve representing their performance in relation to a white line representing their target force of 10% MVC. We then reduced the fidelity of the feedback from a force trace and target force line to a red or green light on screen. This ensured the participants were not heavily reliant on visual feedback to complete the pinch task. Following the training session, participants progressed through a baseline condition followed by three randomized and counterbalanced breathing frequency conditions. For the experimental baseline and breathing conditions, participants did not receive any feedback following each trial. Abbreviated instructions for the breathing conditions are in Table 3. Each of the three breathing conditions consisted of 10 trials completed free from experimenter interaction in a sound-attenuated room. Prior to each of the breathing conditions, participants breathed with a metronome for two minutes to acclimate to the selected breathing frequency, and then continued to breathe with the metronome while performing 10 pinch task trials.

The 10 trials within each condition consisted of an image slideshow of randomized and counterbalanced neutral images of faces, varying in gender and age, across conditions. Neutral images served as attentional control in preparation for the pinch task cue (Coan & Allen, 2007). Images within each condition included neutral face images while images used during the training sessions included neutral images of scenery, objects, and people. Pictures were selected according to normative ratings (NIMH, CSEA: Lang, Bradley, & Cuthbert, 2008) from the International Affective Picture System (IAPS) and internet search engines (e.g. Google, Pixabay, Yahoo). Example of a neutral image is included in experimental protocol figure (see Fig. 2). Experimental conditions consisted of the

Table 2

Breathing frequency data summarizes averages for set and observed breathing paces for all participants across each breathing condition. Means (*M*) and standard deviations (*SD*) are presented for each breathing condition.

Breathing condition	Set pace	Observed pace
	<i>M</i> (<i>SD</i>)	<i>M</i> (<i>SD</i>)
Baseline	NA	14.81 (4.30)
Normal	14.94 (3.96)	14.33 (3.68)
Slow	5.38 (0.75)	5.89 (1.64)
Fast	24.91 (7.53)	23.89 (6.95)

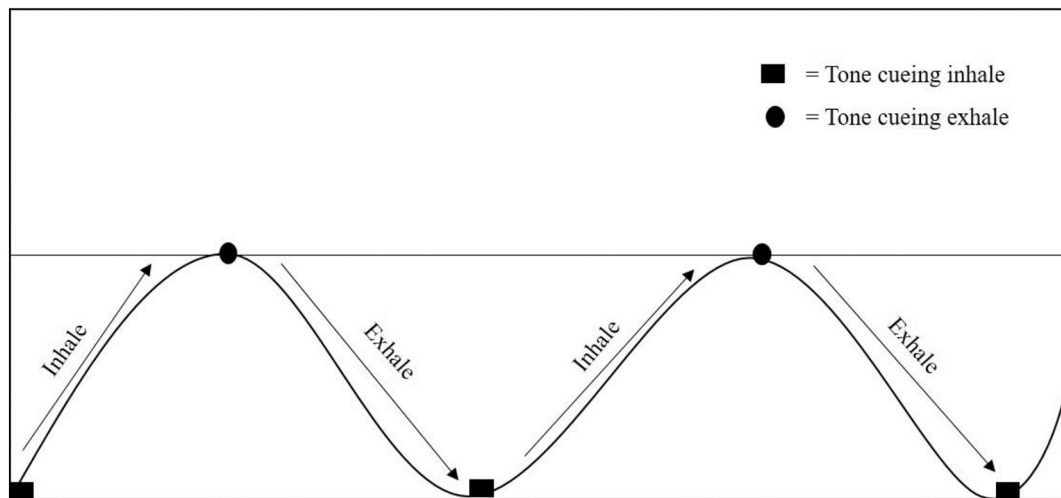


Fig. 1. Graphical representation of metronome pacing accuracy. Squares represent a tone cueing participant to inhale, and circles represent a tone cueing participant to exhale. The sine wave represents inhaling and exhaling as continuous breaths.

Table 3

Abbreviated condition instructions.

Baseline:

...Maintain attention on the image for the entire time it is presented. Once the image disappears and the pixelated image appears, respond by pinching the force transducer with a force equal to 10% of your maximum pinch force. You will complete this 10% pinch as quickly and accurately as possible...

Breathing condition:

... At the end of the two-minute mark, you will be presented with a slide show of images. Please continue to breath in sync with the sound as the images are presented...

presentation of a fixation cross (2 s) upon a black background, image presentation (5–8 s), and completion of the targeted pinch task upon onset image pixelation (2.5 s). The fixation cross directed participants' visual attention to the screen to prepare for the pinch task cue. Presentation of neutral images controlled valence and arousal experienced prior to image presentation cue, as well as provide enough time to record HRV measures prior to pinch task. We tightly controlled image content, to attribute HRV and visual analog scale to changes in breathing frequency. Pixelated images served as a washout period for image cues (see Fig. 3). Following each trial within the training sessions, participants received visual performance feedback. Experimental baseline and breathing conditions did not include pinch task feedback. Feedback consisted of a white horizontal line, representing the 10% target force, and a green or red force trace that represented the force produced.

2.5. Subjective outcomes

2.5.1. Affect grid

The Affect Grid quantifies subjective valence and arousal levels using a 9×9 grid (Russell, Weiss, & Mendelsohn, 1989). The horizontal axis represents valence: 1 (unpleasant) to 9 (pleasant). The vertical axis represents arousal: 1 (low) to 9 (high). We selected this scale to simultaneously measure the dimensional properties of emotion in response to each breathing condition. Following experimental conditions, participants rated overall valence and arousal experienced during breathing and the pinch task.

2.5.2. Visual analog scales

We provided visual analog scales for hindrance, and dyspnea (shortness of breath) experienced following each breathing condition. Participants were asked to mark a vertical dash on a horizontal line based on how they felt breathing impacted performance on the pinch task, numbered 1 (did not hinder; did not facilitate; not noticeable/ not unpleasant) to 10 (did hinder; did facilitate; maximal imaginable intensity/unpleasantness). Scales determine if individual's expectations matched objective outcomes of performance.

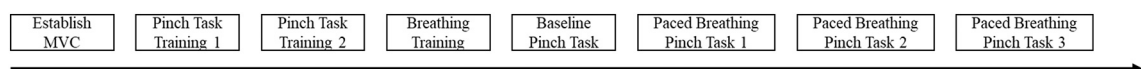


Fig. 2. Experimental protocol. MVC: Maximum Voluntary Contraction.



Fig. 3. Single trial of the pinch task with image presentation.

2.6. Objective outcomes

2.6.1. HRV manipulation

Breathing frequency regulates underlying mechanisms of physiological arousal, including heart rate variability. HRV is the complexity of the timing between heartbeats and is indicative of regulation of autonomic nervous system activity (Shaffer & Ginsberg, 2017). Shorter time between peaks of heart waves are associated with decreased HRV, whereas longer time between peaks of heart waves are associated with increased HRV (Quintana & Heathers, 2014). Increases in HRV characterize adaptive emotional states, decreased subjective arousal, and efficient performance with long-term benefits to emotional and physiological health. In contrast, decreases in HRV characterize maladaptive emotional states, increased subjective arousal, and inefficient performance with long-term consequences to emotional and physiological health (Murray & Raedeker, 2008; Prinsloo et al., 2011; Williams et al., 2015). Slow breathing frequencies (~4.5-6 bpm) increase HRV, while fast or abnormally slow breathing frequencies (≤ 3 bpm) decrease HRV (Song & Lehrer, 2003). Standard deviation of “NN” intervals (SDNN: the standard deviation for “normal” excluding abnormal heart beats) indexed HRV, given that it is an index of timing variability associated with parasympathetic arousal (Levin & Swoap, 2019). It can be calculated by taking the square root of the variance for the duration of NN intervals ($\sqrt{1/N-1 \sum RR_n - \bar{RR}}^2$). In short-term paced breathing protocols, SDNN is most sensitive to parasympathetically mediated processes, compared to other time-domain indices and frequency indices, which is why we focused on this measure only. RR intervals were recorded via the Polar heart rate monitor during each breathing condition. Using Kubios Software, recordings were visually inspected for noise and corrected using the software’s built-in filter function using a Very Low threshold filter to remove artefacts. We did not correct for respiratory frequency as previous studies have demonstrated RSA is not dependent on spontaneous or controlled respiration frequency (Denver, Reed, & Porges, 2007; Laborde, Allen, Göhring, & Dosseville, 2017; Thayer, Loerbroks, & Sternberg, 2011). However, when breathing at slow controlled frequencies changes in frequency-domain power switches from HF to LF which vagal activity is unlikely to be accurately interpreted, as well as the time-domain measure RMSSD. For these reasons, we provide frequency measures and RMSSD measures in Table 5 for descriptive purposes.

2.6.2. Force outcomes and data processing

Force data were processed offline using a customized LabVIEW program (14.0.1). Dependent measures included AE, CE, V, NMT and RT. AE was calculated by subtracting each trial’s peak force from the target force value, squaring the difference, calculating the mean of the squared differences, then computing the square root of each mean value. AE scores represented total error on the task controlling for error direction (above or below the target). CE was calculated as the directional bias of the peak force from the target force value in each trial. Target value was quantified as 10% MVC determined at the beginning of the experiment during the pinch task training. V was calculated by detrending each trial (force onset to peak force) and dividing the standard deviation of the detrended data by the mean force value. Specifically, V was calculated as the absolute trial-to-trial variability for relevant time points between force onset to peak force. Relative change in the trial-to-trial variability was then estimated by dividing the variability in the standard deviation by the average absolute force within the time of force onset to peak force. PRF was calculated by obtaining the maximum of the first derivative of force between force onset and peak force. NMT was calculated for each trial by dividing the value of the target force by the linear slope of force production (from force onset to peak force) and then multiplying this value by 100 to convert the sampling rate into milliseconds. RT was calculated by measuring the time between picture offset and the onset of force production. A customized LabVIEW program identified picture offset frame and identified peak force value to create an array. From the array values, force onset was measured as the time point when force production increased over a threshold of 6 Newtons force output produced during the 100 ms prior to image offset. RTs less than 100 ms following picture offset were deemed “anticipation trials” and removed by macros in excel and visual inspection. Additionally, trials with RTs or raw error scores ± 3 standard deviations from the mean were removed. Forty-three trials were removed due to missing data for variability, and one trial was removed due to an outlier value deemed an anticipation trial for reaction time. Following exclusions, 97.66% of all trials were included in the analyses.

2.7. Statistical analyses

Kolmogorov-Smirnov tests revealed abnormally distributed data. Therefore, we performed non-parametric tests. We conducted Friedman tests for one-way repeated measures to analyze each dependent measure across the change scores of three conditions from baseline to test experimental hypotheses concerning performance differences for each dependent measure. Change scores controlled for motor task and metronome task variation. Change scores calculated the difference in the raw score in each breathing condition from the baseline condition. Significance probability values were initially set at $p < 0.05$. However, to control for multiple comparisons Bonferroni corrections were manually applied with a new significance probability value set at $p < 0.017$ for all analyses. For significant effects, Wilcoxon signed-rank tests were conducted. Results are presented as z scores and significance values.

3. Results

Summary tables outlines all means, standard deviations, uncorrected p values, and corrected p values for questionnaires (see Table 4), HRV (see Table 5), and force outcomes (see Table 6). Following below are results for each dependent variable, and detailed graphs for significant variables.

3.1. Manipulation check

Analysis of SDNN revealed a significant difference of HRV $\chi^2(2) = 82.09, p < 0.001$, for breathing conditions. All conditions were significantly different from each other. Pairwise comparisons identified greater HRV for slow breathing compared to normal ($Z = -5.97, p < 0.001, r = -0.871$) and fast ($Z = -5.97, p < 0.001, r = -0.871$) breathing conditions. Fast breathing conditions had lesser HRV compared to normal and slow breathing ($Z = -4.71, p < 0.001, r = -0.687$). Fig. 4 displays change scores for SDNN for each breathing condition.

Analysis of HR revealed a significant difference $\chi^2(3) = 44.44, p < 0.001$, for breathing conditions. Pairwise comparisons identified greater HR for slow breathing compared to baseline ($Z = -4.06, p < 0.001, r = -0.592$) and lesser HR compared to fast ($Z = -2.51, p = 0.012, r = -0.366$) breathing conditions. Controlling for the metronome with change scores and multiple comparisons, analysis of HR revealed a non-significant difference $\chi^2(2) = 7.36, p = 0.025$, for breathing conditions.

Analysis of mean RR intervals revealed a significant difference $\chi^2(3) = 38.90, p < 0.001$, for breathing conditions. Pairwise comparisons identified lesser mean RR intervals for slow ($Z = -4.14, p < 0.001, r = -0.604$), normal ($Z = -4.93, p < 0.001, r = -0.719$), and fast ($Z = -5.38, p < 0.001, r = -0.785$) breathing compared to baseline. Controlling for the metronome with change scores and multiple comparisons, analysis of mean RR intervals revealed a non-significant difference $\chi^2(2) = 4.63, p = 0.099$, for breathing conditions.

Analysis of RMSSD revealed a significant difference $\chi^2(3) = 26.26, p < 0.001$, for breathing conditions. Pairwise comparisons identified greater RMSSD for slow breathing ($Z = -4.40, p < 0.001, r = -0.642$) compared to baseline. Controlling for the metronome with change scores and multiple comparisons, analysis of RMSSD revealed a significant difference $\chi^2(2) = 18.17, p < 0.001$, for breathing conditions. Fast breathing significantly decreased RMSSD compared to normal ($Z = -3.09, p = 0.002, r = -0.451$) and slow ($Z = -3.69, p < 0.001, r = -0.538$) breathing conditions. Normal and slow were not significantly different from each other.

Analysis of LF-HRV revealed a significant difference $\chi^2(3) = 88.58, p < 0.001$, for breathing conditions. Pairwise comparisons identified greater LF-HRV for slow breathing compared to baseline ($Z = -5.97, p < 0.001, r = -0.871$), fast ($Z = -5.97, p < 0.001, r = -0.871$), and normal ($Z = -5.97, p < 0.001, r = -0.871$) conditions. Controlling for the metronome with change scores and multiple comparisons, analysis of LF-HRV revealed a significant difference $\chi^2(2) = 71.02, p < 0.001$, for breathing conditions. Slow breathing significantly increased LF-HRV compared to fast ($Z = -5.97, p < 0.001, r = -0.871$) and normal ($Z = -5.97, p < 0.001, r = -0.871$) conditions.

Analysis of HF-HRV revealed a significant difference $\chi^2(3) = 42.01, p < 0.001$, for breathing conditions. Pairwise comparisons identified greater HF-HRV for normal breathing compared to baseline ($Z = -3.51, p < 0.001, r = -0.512$), fast ($Z = -5.32, p < 0.001, r = -0.776$), and slow ($Z = -4.37, p < 0.001, r = -0.637$) conditions. Controlling for the metronome with change scores and multiple comparisons, analysis of HF-HRV revealed a significant difference $\chi^2(2) = 37.83, p < 0.001$, for breathing conditions. Normal breathing significantly increased HF-HRV compared to fast ($Z = -5.32, p < 0.001, r = -0.776$) and slow ($Z = -4.37, p < 0.001, r = -0.637$) conditions.

Analysis of Total Power revealed a significant difference $\chi^2(3) = 101.48, p < 0.001$, for breathing conditions. Pairwise comparisons identified greater Total Power for slow breathing compared to baseline ($Z = -5.97, p < 0.001, r = -0.871$), normal ($Z = -5.97, p < 0.001, r = -0.871$), and fast ($Z = -5.97, p < 0.001, r = -0.871$) conditions. Controlling for the metronome with change scores and multiple comparisons, analysis of Total Power revealed a significant difference $\chi^2(2) = 83.53, p < 0.001$, for breathing conditions.

Table 4

Variable data summarizes the dependent variables for all participants across each breathing condition. Means (M) and standard deviations (SD) are presented for each dependent variable: Arousal, Hindrance and Dyspnea as Likert Scales (1–9).

Condition	Variable	Mean (SD)	p -value	
			Baseline	Change score
Baseline	Arousal	4.34 (1.51)	–	–
	Hindrance	2.34 (2.37)	–	–
	Dyspnea	1.45 (1.71)	–	–
Normal	Arousal	4.66 (1.83)	0.200	–
	Hindrance	5.40 (2.56)	0.001*	–
	Dyspnea	2.63 (2.31)	0.001*	–
Slow	Arousal	4.49 (1.65)	0.773	0.243
	Hindrance	4.65 (2.78)	0.001*	0.082
	Dyspnea	2.32 (2.45)	0.004	0.187
Fast	Arousal	5.43 (2.02)	0.002*	0.004*
	Hindrance	6.35 (2.12)	0.001*	0.023
	Dyspnea	4.13 (2.39)	0.001*	0.001*

Table 5

Variable data summarizes the dependent variables for all participants across each breathing condition. Means (*M*) and standard deviations (*SD*) are presented for each dependent variable: Standard Deviation of the RR intervals (SDNN) in milliseconds (ms), Root-mean-succesive differences (RMSSD) in milliseconds (ms), Low frequency (LF), High frequency (HF), and Heart Rate Variability (HRV).

Condition	Variable	Mean (SD)	p-value	
			Baseline	Change score
Baseline	SDNN (ms)	58.56 (25.04)	–	–
	RMSSD (ms)	59.76 (34.02)	–	–
	Mean RR interval (ms)	903.17 (131.47)	–	–
	HR (bpm)	67.74 (9.46)	–	–
	LF-HRV (ms2)	1796.81 (1518.13)	–	–
	HF-HRV (ms2)	1666.30 (1801.79)	–	–
	Total HRV (ms2)	3692.51 (3100.97)	–	–
	LF/HF ratio	1.83 (2.15)	–	–
Normal	SDNN (ms)	64.31 (29.87)	0.146	–
	RMSSD (ms)	69.21 (40.88)	0.017	–
	Mean RR interval (ms)	849.64 (127.60)	0.001*	–
	HR (bpm)	72.15 (10.72)	0.001*	–
	LF-HRV (ms2)	1627.11 (1489.58)	0.101	–
	HF-HRV (ms2)	2920.53 (3305.73)	0.001*	–
	Total HRV (ms2)	4734.26 (4453.45)	0.112	–
	LF/HF ratio	0.95 (0.92)	0.001*	–
Slow	SDNN (ms)	108.77 (32.93)	0.001*	0.001*
	RMSSD (ms)	72.44 (31.48)	0.001*	0.155
	Mean RR interval (ms)	860.15 (112.52)	0.001*	0.099
	HR (bpm)	70.83 (8.91)	0.001*	0.050
	LF-HRV (ms2)	10,858.47 (6428.40)	0.001*	0.001*
	HF-HRV (ms2)	1419.36 (1419.04)	0.059	0.001*
	Total HRV (ms2)	12,489.91 (7535.62)	0.001*	0.001*
	LF/HF ratio	12.39 (8.55)	0.001*	0.001*
Fast	SDNN (ms)	51.73 (21.96)	0.003*	0.001*
	RMSSD (ms)	60.86 (33.38)	0.866	0.002*
	Mean RR interval (ms)	843.62 (127.65)	0.001*	0.537
	HR (bpm)	72.87 (10.71)	0.001*	0.190
	LF-HRV (ms2)	1315.17 (1216.35)	0.019	0.121
	HF-HRV (ms2)	1189.49 (1666.29)	0.001*	0.001*
	Total HRV (ms2)	2678.96 (2442.49)	0.001*	0.001*
	LF/HF ratio	2.26 (1.92)	0.210	0.001*

Slow breathing significantly increased Total Power compared to normal ($Z = -5.97, p < 0.001, r = -0.871$) and fast ($Z = -5.97, p < 0.001, r = -0.871$) conditions.

Analysis of LF-HF ratio revealed a significant difference $\chi^2(3) = 97.75, p < 0.001$, for breathing conditions. Pairwise comparisons identified greater LF-HF ratio for slow breathing compared to baseline ($Z = -4.28, p < 0.001, r = -0.624$), normal ($Z = -5.97, p < 0.001, r = -0.871$), and fast ($Z = -5.96, p < 0.001, r = -0.869$) conditions. Controlling for the metronome with change scores and multiple comparisons, analysis of LF-HF ratio revealed a significant difference $\chi^2(2) = 77.11, p < 0.001$, for breathing conditions. Slow breathing significantly increased LF-HF ratio compared to normal ($Z = -5.97, p < 0.001, r = -0.871$) and fast ($Z = -5.96, p < 0.001, r = -0.869$) conditions.

3.2. Accuracy (AE, CE)

AE significantly differed depending on which type of breathing frequency was used, $\chi^2(3) = 50.46, p < 0.001$. Post hoc analysis with Wilcoxon signed-rank test showed that fast ($Z = -7.69, p = 0.001, r = -0.253$), slow ($Z = -5.64, p = 0.011, r = -0.187$) and normal ($Z = -7.11, p < 0.001, r = -0.234$) breathing frequencies increased AE compared to baseline breathing frequencies. When controlling for metronome differences, there were no statistically significant differences in AE between breathing frequency conditions $\chi^2(2) = 4.90, p = 0.086$.

CE significantly differed depending on which type of breathing frequency was used, $\chi^2(3) = 49.07, p < 0.001$. Post hoc analysis with Wilcoxon signed-rank test showed that fast ($Z = -6.64, p = 0.001, r = -0.218$), slow ($Z = -5.69, p = 0.011, r = -0.189$) and normal ($Z = -6.49, p < 0.001, r = -0.214$) breathing frequencies increased CE compared to baseline breathing frequencies indicating participants overshot the target. When controlling for metronome differences, there were no statistically significant differences in CE between breathing frequency conditions $\chi^2(2) = 1.39, p > 0.05$.

3.3. Variability (V)

V significantly differed depending on which type of breathing frequency was used, $\chi^2(3) = 28.30, p < 0.001$. Post hoc analysis with Wilcoxon signed-rank test showed that fast ($Z = -3.30, p = 0.001, r = -0.108$), slow ($Z = -2.53, p = 0.011, r = -0.084$) and normal (Z

Table 6

Variable data summarizes the dependent variables for all participants across each breathing condition. Means (*M*) and standard deviations (*SD*) are presented for each dependent variable: Absolute endpoint error (AE) and Constant Error (CE) in Newtons (N), Trial-to-trial Variation of the Force (V) as a percentage of force (%), Reaction Time (RT) in ms, Normalized Movement Time (NMT) in ms, and Peak Rate of Force as N per second (N/s).

Condition	Variable	Mean (SD)	<i>p</i> -value	
			Baseline	Change score
Baseline	AE (N)	9.48 (29.92)	–	–
	CE (N)	1.4327 2.84217	–	–
	V (%)	10.59 (6.81)	–	–
	RT (ms)	725.57 (300.57)	–	–
	NMT (ms)	312.55 (202.85)	–	–
	PRF (N/s)	48.25 (29.11)	–	–
Normal	AE (N)	24.56 (56.56)	0.001*	–
	CE (N)	2.55 (4.18)	0.001*	–
	V (%)	12.06 (7.93)	0.001*	–
	RT (ms)	729.11 (268.46)	0.487	–
	NMT (ms)	303.28 (225.36)	0.001*	–
	PRF (N/s)	57.55 (36.64)	0.001*	–
Slow	AE (N)	25.76 (71.27)	0.001*	0.015*
	CE (N)	2.59 (4.34)	0.001*	0.353
	V (%)	11.42 (7.59)	0.011*	0.032
	RT (ms)	783.61 (339.63)	0.001*	0.001*
	NMT (ms)	305.17 (257.04)	0.009*	0.757
	PRF (N/s)	57.60 (41.79)	0.001*	0.267
Fast	AE (N)	24.27 (47.32)	0.001*	0.726
	CE (N)	2.65 (4.15)	0.001*	0.565
	V (%)	12.23 (9.19)	0.001*	0.368
	RT (ms)	694.10 (266.68)	0.130	0.013*
	NMT (ms)	281.83 (226.50)	0.001*	0.001*
	PRF (N/s)	59.82 (34.12)	0.001*	0.097

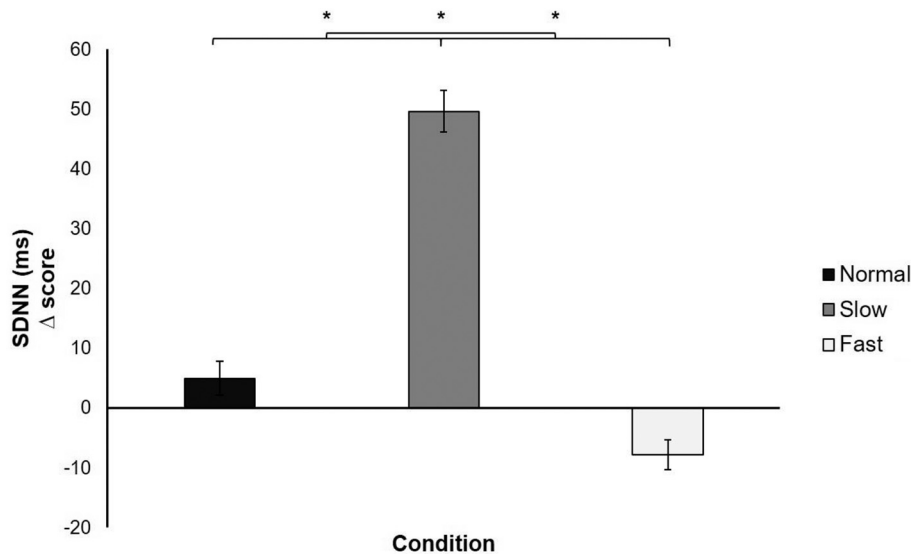


Fig. 4. SDNN across each breathing frequency condition. Error bars represent ± 1 standard error (SE) from the mean. SDNNs are change scores from each breathing condition from baseline in milliseconds (ms). Values above the origin indicate an increase in HRV in comparison to the baseline condition. Asterisks represent significant differences ($p < 0.05$).

$= -4.64, p < 0.001, r = -0.153$) breathing frequencies increased V compared to baseline breathing frequencies. When controlling for metronome differences, there were no statistically significant differences in V between breathing frequency conditions $\chi^2(2) = 4.94, p = 0.084$.

3.4. Speed (RT, MT, PRF)

RT significantly differed depending on which type of breathing frequency was used, $\chi^2(3) = 23.24, p < 0.001$. Post hoc analysis

with Wilcoxon signed-rank tests showed that slow breathing frequencies increased reaction time compared to baseline ($Z = -3.80, p < 0.001, r = -0.187$). Fast, normal, and baseline did not significantly differ from each other. Controlling for metronome effects with change scores, RT significantly differed depending on which type of breathing frequency was used, $\chi^2(2) = 24.10, p < 0.001$. Post hoc analysis with Wilcoxon signed-rank test showed that fast breathing frequencies decreased reaction time compared to slow ($Z = -5.41, p < 0.001, r = -0.179$) and normal ($Z = -2.48, p = 0.013, r = -0.082$) breathing frequencies. Slow breathing frequencies increased RT compared to normal ($Z = -3.83, p < 0.001, r = -0.127$) breathing frequencies. Fig. 5 displays change scores for RT for each breathing condition.

NMT significantly differed depending on which type of breathing frequency was used, $\chi^2(3) = 53.38, p < 0.001$. Post hoc analysis with Wilcoxon signed-rank test showed that fast ($Z = -7.02, p < 0.001, r = -0.231$), slow ($Z = -2.60, p = 0.009, r = -0.086$) and normal ($Z = -3.63, p < 0.001, r = -0.119$) breathing frequencies significantly decreased NMT compared to baseline breathing frequencies. However, normal and slow breathing NMT did not significantly differ NMT significantly differed depending on which type of breathing frequency was used, $\chi^2(2) = 23.87, p < 0.001$. Post hoc analysis with Wilcoxon signed-rank test showed that fast breathing frequencies decreased movement time compared to slow ($Z = -4.74, p < 0.001, r = -0.157$) and normal ($Z = -4.48, p < 0.001, r = -0.147$) breathing frequencies. Slow breathing frequencies did not significantly impact NMT compared to normal breathing frequencies. Fig. 6 displays change scores for MT for each breathing condition.

PRF significantly differed depending on which type of breathing frequency was used, $\chi^2(3) = 28.30, p < 0.001$. Post hoc analysis with Wilcoxon signed-rank test showed that fast ($Z = -8.87, p < 0.001, r = -0.291$), slow ($Z = -5.94, p < 0.001, r = -0.197$) and normal ($Z = -7.23, p < 0.001, r = -0.238$) breathing frequencies increased PRF compared to baseline breathing frequencies. When controlling for metronome effects, there were no statistically significant differences in PRF between breathing frequency conditions $\chi^2(2) = 8.17, p = 0.017$, based on Bonferroni corrections for multiple comparisons.

3.5. Arousal

Analysis of arousal revealed a significant main effect of breathing condition $\chi^2(3) = 12.11, p < 0.01$. Pairwise comparisons identified a significant difference of increased arousal for fast breathing compared to baseline ($Z = -3.10, p = 0.002, r = 0.452$), slow ($Z = -3.36, p < 0.001, r = 0.490$) and normal ($Z = -2.87, p = 0.004, r = 0.419$) breathing conditions. Analysis of arousal revealed a significant main effect of breathing condition $\chi^2(2) = 6.47, p < 0.05$. Pairwise comparisons identified a significant difference of increased arousal for fast breathing compared to slow ($Z = -3.36, p < 0.001, r = 0.490$) and normal ($Z = -2.87, p = 0.004, r = 0.419$) breathing conditions. Fig. 7 displays change scores for arousal for each breathing condition.

A Pearson product-moment correlation was run to determine the relationship between arousal and reaction time by breathing condition. There was a low, negative correlation between arousal and RT, which was statistically significant ($r = -0.291, n = 47, p = 0.047$). Specifically, this relationship was significant for normal metronome-paced breathing ($r = -0.325, n = 47, p = 0.026$).

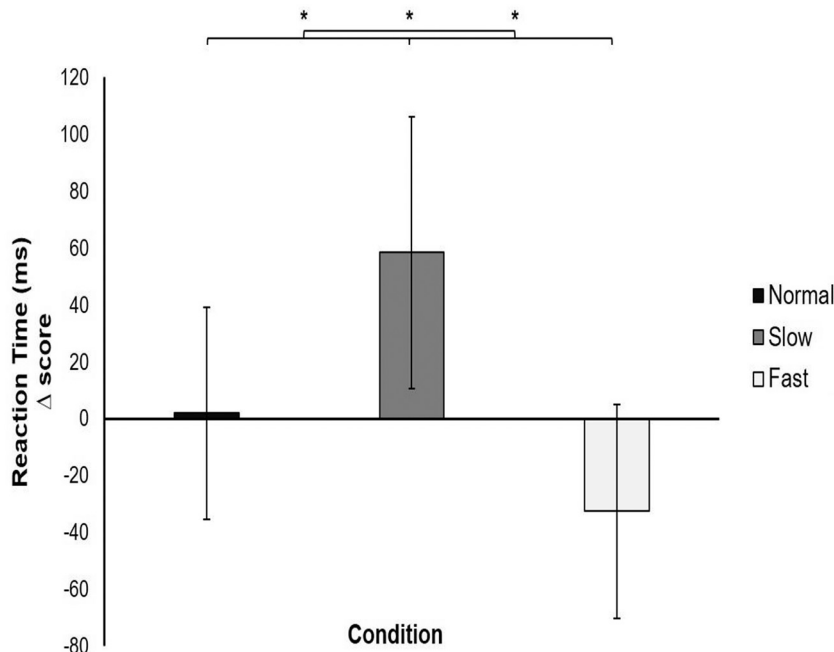


Fig. 5. RT, as a change score from baseline, for each breathing frequency condition in milliseconds (ms). In comparison to the baseline condition, values above the origin indicate increases in time to initiate the pinch task, while values below the origin indicate decreases in time to initiate the pinch task. Error bars represent ± 1 SE. Asterisks represent significant differences ($p < 0.05$).

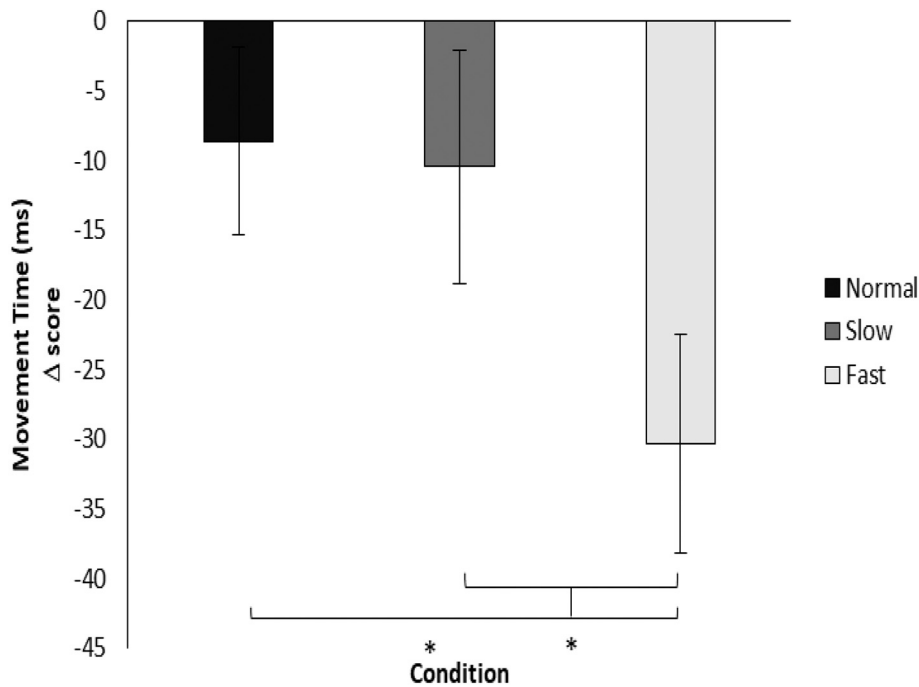


Fig. 6. NMT, as a change score from baseline, for each breathing frequency condition in milliseconds (ms). In comparison to the baseline condition, values above the origin indicate increases in time to perform the pinch task, while values below the origin indicate decreases in time to perform the pinch task. Error bars represent ± 1 SE. Asterisks represent significant differences ($p < 0.05$).

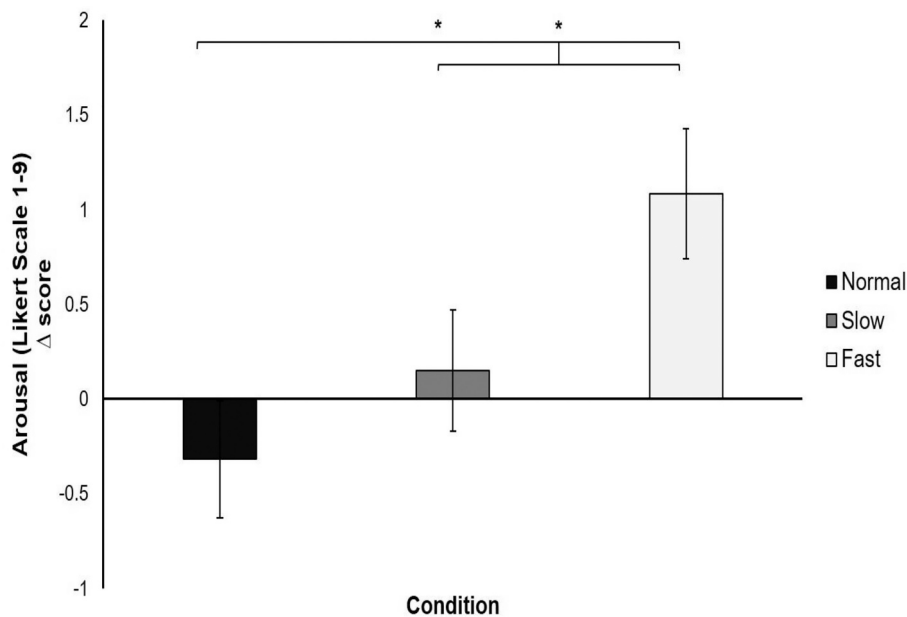


Fig. 7. Arousal, as a change score from baseline, for each breathing frequency condition on a 9-point Likert scale. In comparison to the baseline condition, values above the origin indicate increases in perceived arousal, while values below the origin indicate decreases in perceived arousal during the pinch task. Error bars represent ± 1 SE from the mean. Asterisks represent significant differences ($p < 0.05$).

3.6. Hindrance

Analysis of hindrance revealed a significant main effect of breathing condition $\chi^2(3) = 50.48, p < 0.001$. Pairwise comparisons identified a significant difference of increased hindrance for slow ($Z = -3.84, p < 0.001, r = -0.560$), normal ($Z = -5.19, p < 0.001, r$

= -0.757), and fast ($Z = -5.59, p < 0.001, r = -0.815$) breathing compared to the baseline condition. Analysis of hindrance revealed a significant main effect of breathing condition $\chi^2(2) = 46.35, p < 0.001$. Pairwise comparisons identified lower feelings of hindrance for slow breathing compared to fast breathing conditions ($Z = -3.68, p < 0.001, r = -0.537$). Both slow and fast breathing conditions had increased feelings of hindrance compared to the normal breathing condition ($Z = -4.92, p < 0.001, r = -0.718$) ($Z = -5.60, p < 0.001, r = -0.817$). Fig. 8 displays change scores for hindrance for each breathing condition.

A Pearson product-moment correlation was run to determine the relationship between hindrance and reaction time by breathing condition. There were no significant correlations between hindrance and RT.

3.7. Dyspnea

Analysis of dyspnea revealed a significant main effect of breathing condition $\chi^2(3) = 42.02, p < 0.001$. Pairwise comparisons identified a significant difference of increased dyspnea for slow ($Z = -2.86, p = 0.004, r = -0.417$), normal ($Z = -3.39, p = 0.001, r = -0.494$), and fast ($Z = -5.39, p < 0.001, r = -0.786$) breathing compared to the baseline condition. Controlling for metronome effects, analysis of dyspnea revealed a significant main effect of breathing condition $\chi^2(2) = 32.32, p < 0.001$. Pairwise comparisons identified decreased feelings of dyspnea for slow breathing compared to fast breathing conditions ($Z = -4.25, p < 0.001, r = -0.620$). Both slow and fast breathing conditions increased dyspnea compared to normal ($Z = -3.51, p < 0.001, r = -0.512$) ($Z = -4.93, p < 0.001, r = -0.719$). Fig. 9 displays change scores for hindrance for each breathing condition.

A Pearson product-moment correlation was run to determine the relationship between dyspnea and reaction time by breathing condition. There were no significant correlations between dyspnea and RT.

4. Discussion

We investigated whether breathing frequency influences execution of a goal-directed accuracy-based motor task. We hypothesized fast breathing would decrease the accuracy and speed with which the task was performed when compared to normal and slow breathing. Our data partially supported our hypotheses, extending the literature via two novel results. First, fast breathing was found to facilitate reaction time and movement time, while slow breathing hindered reaction time compared to normal breathing conditions. Second, fast breathing was perceived as being more hindering, arousing, and dyspnea inducing compared to slow and normal breathing conditions. Breathing frequency did not influence accuracy or variability. Our findings provide strong evidence that increasing breathing frequency can facilitate reaction time and movement time, while preserving accuracy. These findings are elaborated in the context of prior work in the broader area of paced breathing and performance.

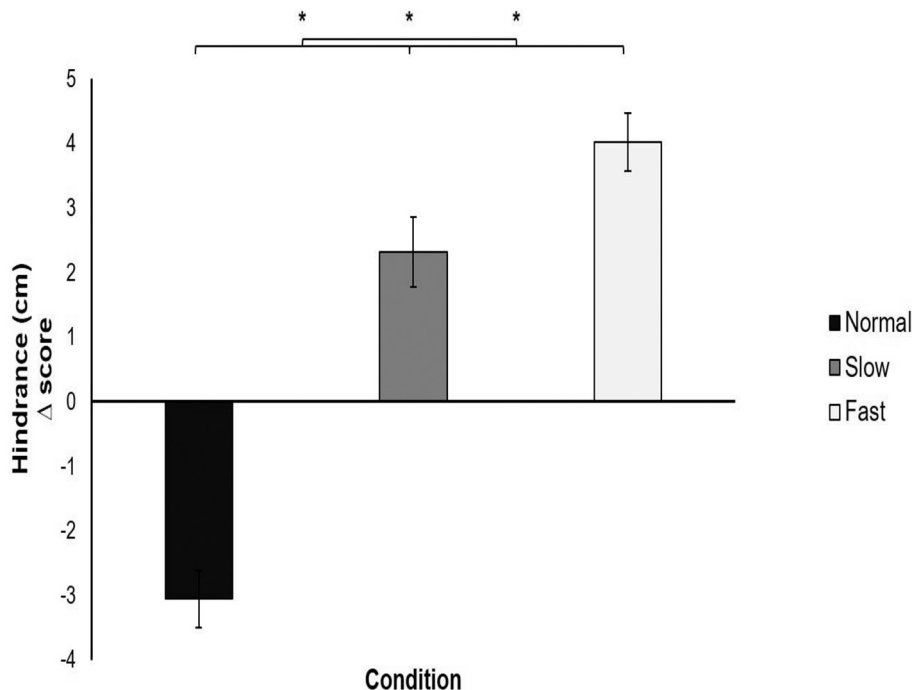


Fig. 8. Hindrance, as a change score from baseline, for each breathing frequency condition in centimeters (cm). In comparison to the baseline condition, values above the origin indicate increases in perceived hindrance of breathing, while values below the origin indicate decreases in perceived hindrance of breathing to the performance of the pinch task. Error bars represent ± 1 SE from the mean. Asterisks represent significant differences ($p < 0.05$).

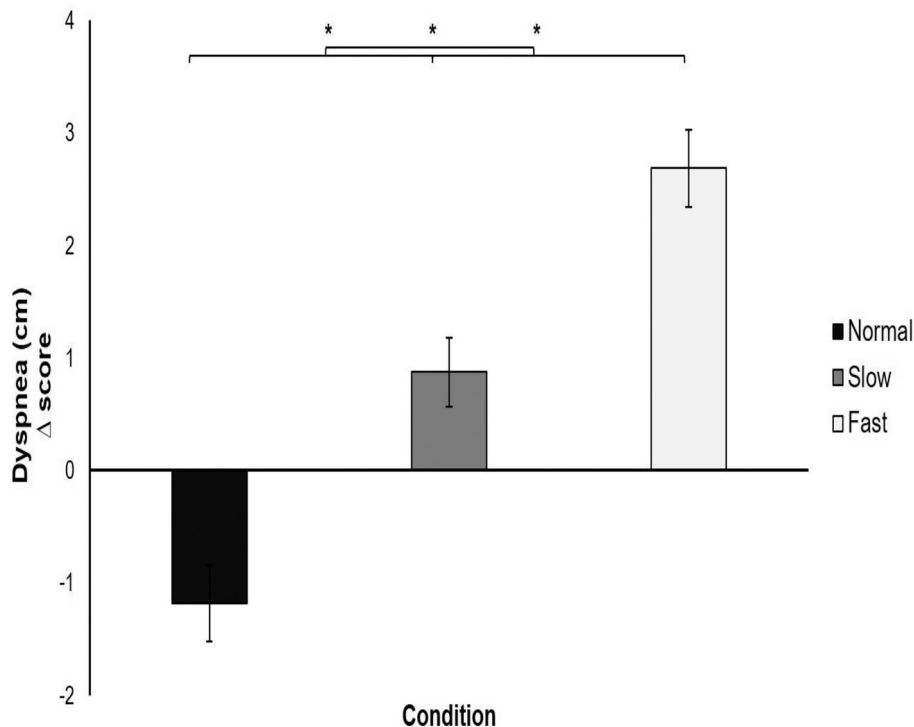


Fig. 9. Dyspnea, as a change score from baseline, for each breathing frequency condition in centimeters (cm). In comparison to the baseline condition, values above the origin indicate increases in perceived shortness of breath, while values below the origin indicate decreases in perceived shortness of breath during the performance of the pinch task. Error bars represent ± 1 SE from the mean. Asterisks represent significant differences ($p < 0.05$).

4.1. Manipulation check

We replicated previous findings that breathing frequency alters HRV and subjective arousal (Martin et al., 2012). Fast breathing decreases HRV, while slow breathing increases HRV (Song & Lehrer, 2003). HRV indexes sympathetic and parasympathetic activation of the autonomic nervous system. SDNN decreased during the fast breathing condition, providing evidence that participants are physiologically aroused when breathing faster than normal. Consistent with previous literature, average HR did not significantly differ by respiratory rate. For frequency response changes, slow breathing was consistent with previous studies showing increases in LF-HRV, Total Power, LF/HF ratio and decreases in HF-HRV compared to normal and fast breathing rates (Guzik et al., 2007; Song & Lehrer, 2003; Van Diest et al., 2014; Zaccaro et al., 2018). Further, we also replicated findings that paced breathing does not impact mean RR intervals and RMSSD (Penttilä et al., 2001; Sandercock et al., 2008). In addition to expected HRV responses, observed breathing rates were similar to targeted breathing rates. Observed increased in physiological arousal matched increases in subjective arousal, hindrance, and dyspnea.

4.2. Breathing frequency and motor planning and execution

We expected accuracy and variability to differ amongst breathing conditions. However, accuracy and variability of the pinch task did not waiver across the various breathing frequencies. Results are not consistent with literature showing improvements in cognitive performance accuracy when breathing slowly (Prinsloo et al., 2011). A potential explanation for our finding lies the natural prioritization of accuracy over speed in execution of the current task based on the instructions given to participants. Opposing changes in speed between fast and slow breathing conditions potentially drove between condition differences in subjective and objective arousal and unpleasant emotional states. Such variable conditions elicit alterations in speed and accuracy prioritization. Previous evidence suggests unpleasant emotional states increase speed of non-directional discrete motor tasks (Coombes, Higgins et al., 2009). When breathing at an accelerated pace, the fast breathing frequency condition increased arousal, hindrance, and dyspnea, indicative of an unpleasant affective state. Therefore, our findings are consistent with the notion that unpleasant emotional states facilitate movement speed. Relative to the extant database, the novelty of the current findings lies in connecting the breathing frequency manipulation with the elicitation of unpleasant emotional states and resulting performance alterations.

Additionally, we observed increases in subjective arousal for fast breathing frequencies compared to slow and normal breathing frequencies. Metronome pacing possibly increased attentional, cognitive, and motor demands. Such an interpretation emerges from evidence of competing attentional resource allocation to internal (i.e. breathing frequency) and external cues (i.e. pinch task) (Busch

et al., 2012). Reaction time and movement time decreased during the fast breathing condition, suggesting that breathing more rapidly enabled greater attentional capacity which was then available and directed to the motor task. Fast breathing is more rhythmic, thereby permitting more attention to the motor task (Herrero, Khuvis, Yeagle, Cerf, & Mehta, 2018). In accord with this rationale, previous evidence shows high frequency breathing interventions improve one's ability to focus and shift attention due to its enhanced rhythmicity (Joshi & Telles, 2009). Rhythmic movement such as swimming is met with natural increases in respiratory rate (Leahy et al., 2019). Increases in respiratory rate are a natural response to increases in task workload (Grassmann, Vlemincx, Von Leupoldt, Mittelstädt, & Van den Bergh, 2016; Mortola, Marghescu, Siegrist-Johnstone, & Matthes, 2020). In contrast, reducing breathing frequency is incompatible with reducing greater motor output such as that which is required for executing similar movements with greater speed. Fast breathing, albeit more uncomfortable, apparently met the task demands of quickly executing the pinch task more effectively than slow breathing. In turn, the slower movement initiation and execution evidenced here during slow breathing is likely attributable to it being more attention demanding.

4.3. Motor benefits and costs of breathing frequency manipulation

Fast, relative to slow and normal breathing, was reported as being more hindering to performance during the pinch task. Homma and Masaoka (2008) suggest faster breathing frequencies accompany maladaptive emotional states (e.g. anxiety) and slower breathing frequencies accompany adaptive emotional states (e.g. relaxation). In the context of emotional health, unpleasant emotional states are maladaptive and inhibitive. However, in the context of motor performance, unpleasant emotional states can be facilitative by increasing movement speed (Coombes, Janelle, & Duley, 2005). The simplicity of the task is a related factor to consider in the context of this interpretation. When performing the simple pinch task, increases in arousal most likely precipitated higher activation levels which in turn resulted in a decrease of RT without any measurable accuracy effects (Van Gemmert & Van Galen, 1997). Our results are consistent with this interpretation. Faster breathing was indeed reported as more unpleasant, while also speeding reaction time.

4.4. Implications and future directions

Our findings indicate that within performance contexts emphasizing accuracy, breathing frequencies alter reaction time induced by arousal levels during controlled breathing. However, motor performance changes following breathing rates from slow to fast breathing frequencies may be due to factors other than changes in arousal, dyspnea, or hindrance. Impacts of breathing frequency in movement contexts support the notion that a speed-accuracy tradeoff is dependent on the movement goal. Movement enables individuals to accurately perform tasks while maintaining appropriate arousal levels. Our evidence indicates that attempting to regulate physiological and subjective arousal states can facilitate or inhibit movement initiation. Accordingly, our results highlight the need to appropriately prescribe breathing interventions to meet the demands of specific contexts. For example, performance environments may benefit from faster reaction times elicited under certain arousal states that are sequelae of alterations in breathing parameters. Other performance environments may necessitate prioritization of movement accuracy, in which the speed-based benefits elicited by specific breathing rates are inconsequential or even detrimental to attainment of performance goals.

Within short-term performance contexts, it is reasonable to assume fast breathing provides minimal cost to motor actions and may even facilitate their execution. Reductions in attentional demands likely explain why voluntary fast breathing frequency is ideal. Usually, negative connotations are associated with fast breathing. For example, fast breathing is considered as an involuntary symptom of poor physiological and psychological well-being (Nardi, Freire, & Zin, 2009; Narkiewicz et al., 2006). Despite these negative context specific connotations, our results suggest that fast breathing is a facilitative short-term performance strategy to initiate movement quickly in acute performance contexts.

Although our data highlight the immediate benefits of fast breathing to movement initiation, it is important to note that increases in breathing frequency increase sympathetic nervous system activity. Fast breathing frequencies mainly regulate arousal state without increasing pleasantness of emotion (Mourya, Mahajan, Singh, & Jain, 2009; Philippot, Chapelle, & Blairy, 2002). Therefore, fast breathing frequencies may be ideal for completing acute motor tasks in emotionally neutral environments. However, in emotionally salient or temporally constrained environments, fast breathing frequencies may compromise motor efficiency. Future studies should determine the impact of breathing frequency on motor performance under emotionally salient or temporally constrained contexts.

While slow breathing frequencies were more costly to performance speed than fast breathing frequencies, it is plausible that slow breathing frequencies are advantageous in other performance contexts. Recent investigations identified faster reaction times implementing slow breathing frequencies when the task emphasizes speed. For example, Dhadse and Fadia (2016) found that while employing a breathing intervention that reduced breathing frequency, reaction times were faster. Thus, slow breathing may be ideal in environments where speed is emphasized.

While novel findings emerged herein, the current results should be interpreted with perspective of limitations and delimitations of the project. Participants' initial assumptions of emotional states associated with fast and slow breathing frequencies potentially influenced the current findings, since some participants verbalized recognition of differences between frequencies. Our manipulation included a breathing rate control using a metronome, which did induce changes relative to baseline and normal-paced breathing conditions. We controlled for metronome effects by using change scores to analyze the various force outcomes. We further controlled for inter-subject variability amongst dependent variables induced by breathing conditions by employing a repeated measures design. It remains unclear whether these same changes would occur if participants breathed at different frequencies *without an external pacer*. In spite of these limitations, the current study provides ample evidence that changes in breathing frequency alter motor performance outcomes.

5. Conclusion

The lion's share of breath regulation research focuses on the long-term health outcomes of regulating arousal via various manipulations of breathing rate and depth. Our data imply immediate impacts on speed of movement based on breathing frequency. More specifically, in emotionally neutral situations, breathing faster than one's normal resting rate benefits performing tasks that emphasize accuracy. Replication and extension of these initial findings is necessary to draw firm conclusions regarding the contexts and circumstances for implementing breathing interventions to optimize motor performance.

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Declaration of Competing Interest

None.

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